Hai Duong's
Curriculum Vitae

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1 Education

- **PhD**, Computer Science, George Mason University, Fairfax, VA 2022–present

  Advisor: ThanhVu Nguyen

- **MS**, Electrical Engineering, Hanoi University of Science and Technology, Hanoi, VN 2019–2021

  Advisor: Quoc-Cuong Nguyen

- **BS**, Electrical Engineering, Hanoi University of Science and Technology, Hanoi, VN 2014–2019

2 Experience

- **Applied Scientist Intern**, Amazon Web Services, Amazon 2026
- **Research Assistant**, ROARS Lab, George Mason University 2022–present

  Developed a verification tool, NeuralSAT P5, ranked 4th in VNN-COMP'23, 2nd in '24 and '25.

  Developed optimization techniques for DNN verification, e.g., stabilization P6, composition P3, unsat proof P4, structural perturbations P2, training for verification P7, etc.

  Published papers at ISSTA, FSE, CAV, NeurIPS, and CVPR.

- **Research Assistant**, BachLe's Lab, University of Melbourne 2021–2022

  Worked on a graph-based source code modeling and explanation technique P8.

- **Research Assistant**, Sensor Lab, Hanoi University of Science and Technology 2019–2021

  Developed a speech enhancement system using graph-based neural beamforming P10.

  Developed a small-footprint keyword spotting system using deformable convolution P9.

3 Awards and Honors

- Highlight Paper Award [P1], CVPR 2026
- Broadening Participation Scholarship Award, CVPR 2026
- Spotlight Paper Award [P3], NeurIPS 2025
- Scholar Award, NeurIPS 2025
- Graduate Travel Award, GMU 2025
- NeuralSAT ranked 2nd, VNN-COMP 2024–2025
- NeuralSAT ranked 4th and received New Participant Award, VNN-COMP 2023
- Graduate Scholarship (Full Tuition), HUST 2019–2021
- Outstanding Undergraduate Award, HUST 2015–2018

4 Publications

4.1 Refereed Conference/Journal Papers (Full Papers)

P1 Hai Duong, Lam Nguyen, Thanh Le, and ThanhVu Nguyen. "Verifying Neural Network Robustness with Dual Perturbations". In: *Computer Vision and Pattern Recognition (CVPR)*. 2026, to appear

Highlight Paper

P2 Hai Duong, Thanh Le, Lam Nguyen, and ThanhVu Nguyen. "Verifying Structural Robustness of Deep Neural Network". In: *Proceedings of the ACM on Software Engineering* 3.Foundations of Software Engineering (FSE) (2026), to appear

P3 Hai Duong, David Shriver, ThanhVu Nguyen, and Matthew Dwyer. "Compositional Neural Network Verification via Assume-Guarantee Reasoning". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2025, to appear

Spotlight Paper

P4 Hai Duong, ThanhVu Nguyen, and Matthew Dwyer. "Generating and Checking DNN Verification Proofs". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2025, to appear

P5 Hai Duong, ThanhVu Nguyen, and Matthew Dwyer. "NeuralSAT: A High-Performance Verification Tool for Deep Neural Networks". In: *Computer Aided Verification (CAV)*. 2025, pages 409–423

P6 Hai Duong, Dong Xu, ThanhVu Nguyen, and Matthew Dwyer. "Harnessing Neuron Stability to Improve DNN Verification". In: *Proceedings of the ACM on Software Engineering* 1.Foundations of Software Engineering (FSE) (2024), pages 859–881

P7 Dong Xu, Nusrat Jahan Mozumder, Hai Duong, and Matthew B Dwyer. "Training for Verification: Increasing Neuron Stability to Scale DNN Verification". In: *International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)*. Springer. 2024, pages 24–44

P8 Thanh-Dat Nguyen, Thanh Le-Cong, Duc-Minh Luong, Hai Duong, Xuan-Bach D Le, David Lo, and Quyet-Thang Huynh. "Ffl: Fine-grained fault localization for student programs via syntactic and semantic reasoning". In: *2022 IEEE International Conference on Software Maintenance and Evolution (ICSME)*. IEEE. 2022, pages 151–162

P9 Huu Binh Nguyen, Hai Duong, Anh Xuan Tran Thi, and Quoc Cuong Nguyen. "Efficient keyword spotting system using deformable convolutional network". In: *IETE Journal of Research* 69.7 (2023), pages 4196–4204

P10 Huu Binh Nguyen, Hai Duong, Tien Dat Bui, Hoang Ngoc Chau, and Quoc Cuong Nguyen. "Multi-channel speech enhancement using a minimum variance distortionless response beamformer based on graph convolutional network". In: *International Journal of Advanced Computer Science and Applications* 13.10 (2022)

4.2 Refereed Conference/Workshop Papers (Short Papers)

S1 Hai Duong and ThanhVu Nguyen. "NeuralSAT: Scaling Constraint Solving for DNN Verification (Competition Contribution)". In: *International Symposium on AI Verification (SAIV)*. Springer. 2025, pages 253–259

S2 ThanhVu Nguyen, KimHao Nguyen, and Hai Duong. "SymInfer: Inferring Numerical Invariants using Symbolic States". In: *International Conference on Software Engineering (ICSE)*. IEEE, 2022, pages 197–201

4.3 Unpublished

U1 Hai Duong, Thanh Le, and ThanhVu Nguyen. "DRL-Guided Smart Branching for Neural Network Verification". In: *Submitted*. 2026

U2 Thanh Le, Hai Duong, ThanhVu Nguyen, and Takeshi Matsumura. "Verca: Verifying Robustness of Constrained-Latent Autoencoders". In: *Submitted*. 2026

U3 Junyu Yin, Lingda Li, Hai Duong, Adolfy Hoisie, and Keren Zhou. "Trace-Driven DL-based Framework for GPU Performance Modeling". In: *Submitted*. 2026

U4 Thanh Le, Hai Duong, ThanhVu Nguyen, and Takeshi Matsumura. "Formal Verification of DNN-based Semantic Communication to Adversarial Noise". In: *Submitted*. 2026

U5 Thanh Le, Hai Duong, Yusheng Ji, ThanhVu Nguyen, and John C.S. Lui. "FGGM: Formal Grey-box Gradient Method for Attacking DRL-based MU-MIMO Scheduler". In: *arXiv*. 2025

U6 Hai Duong, ThanhVu Nguyen, and Matthew Dwyer. "A DPLL(T) Framework for Verifying Deep Neural Networks". In: *arXiv*. 2024

5 Miscellaneous

- Contribute to proposals:
  - Amazon Research Award (Build on Trainium: Responsible AI): Leveraging AWS Trainium for Verifiable AI and ML-Assisted Mathematical Reasoning (2026). $250,000 Training Credits. Amazon.
  - NVIDIA Academic Grant Program: Trustworthy AI: Bringing Scalability and Assurance to DNN Verification (2025). DGX Spark System. NVIDIA.
  - Amazon Research Award (Automated Reasoning): Scalable and Precise DNN Constraint Solving with Abstraction and Conflict Clause Learning (2023). $50,000 unrestricted gift. Amazon.
  - NSF CAREER: NeuralSAT: A Constraint-Solving Framework for Verifying Deep Neural Networks (2023). NSF 2238133. 8/1/2023–7/31/2028, $510,509. NSF.
- Help review conference papers:
  - ISSTA'23
  - PLDI'24, OOPSLA'24
  - ICLR'25
  - CVPR'26, ICML'26, ECCV'26, NeurIPS'26

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